

Photosynthetic and growth responses to different light regimes of the major canopy species of Tasmanian cool temperate rainforest

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Abstract

Seedlings of the cool temperate rainforest tree species, *Nothofagus cunninghamii*, *Atherosperma moschatum*, *Eucryphia lucida*, *Phyllocladus aspleniifolius*, *Lagarostrobos franklinii* and *Athrotaxis selaginoides*, were grown in various light regimes and their instantaneous light-dependence curves and relative growth rates were measured. *A. moschatum* had the highest relative growth rate of all species in heavy shade, and its light-dependence curves were also characteristic of a relatively shade-tolerant species. *A. selaginoides* and *L. franklinii* seedlings raised in heavy shade had low instantaneous light compensation points but their growth rates were not distinct from that of *N. cunninghamii*. The maximum relative growth rate of the gymnosperms was low due to a low specific leaf area and leaf area ratio, although the maximum rates of photosynthesis on a leaf area basis were similar to *N. cunninghamii* and *E. lucida*. The low leaf area ratio and specific leaf area affected both the relative growth rate and the growth light compensation point. Although in the heaviest shade treatment there was no difference in growth rate between *N. cunninghamii*, *E. lucida*, *L. franklinii* and *A. selaginoides*, in moderate shade treatments *N. cunninghamii* had a higher relative growth rate than the gymnosperms and a higher maximum rate of photosynthesis on a leaf dry weight basis than all species when raised in semi-shaded natural light conditions. These species commonly occur in mixture in Tasmanian cool temperate rainforest and these results are discussed in terms of forest dynamics.

Introduction

Nothofagus species dominate much of the

southern hemisphere cool temperate rainforest. The cool temperate rainforest of Chile is dominated by *Nothofagus* species at high latitudes and altitudes. *Nothofagus* species also dominate some low altitude and mid-elevation sites in south-central Chile (39–42°S) which are exposed to recurrent catastrophic disturbance (Veblen *et al.* 1981). Similarly in New Zealand *Nothofagus* may dominate on sites of recent forest disturbance, but generally increases in dominance away from the moist, mild, fertile optimum (Wardle 1983, 1984). Wardle (1984) suggested that *Nothofagus* seedlings are better able to tolerate harsh conditions than most other rainforest species but are poor competitors. In both Chile and New Zealand the poor competitive ability on the more equable sites (moist and mild on fertile soils) has been related at least in part to higher light requirements of the *Nothofagus* species relative to co-occurring canopy species (Veblen *et al.* 1981; Wardle 1970).

The most striking contrast between the Australian cool temperate rainforests and those of Chile and New Zealand, lies in the dominance by *Nothofagus cunninghamii* of apparently climax forest on equable sites in south-eastern Australia. Cool temperate rainforest dominated by *N. cunninghamii* occurs in south-central Victoria and in Tasmania as lowland and montane communities. This forest is most widespread in Tasmania, occurring over a range of soils and topographies (Jackson 1968) and has a greater diversity of floristic associations and forest structures than in Victoria (Busby & Bridgewater 1977). Lowland rainforest on fertile soils forms a tall closed forest (Specht 1970) with open understorey, dominated by *N. cunninghamii* with *Atherosperma moschatum* (Monimiaceae) as sub-dominant. On the less fertile soils of western Tasmania, these species form mixtures with *Eucryphia lucida* (Eucryphiaceae) and *Phyllocladus aspleniifolius* (Podocarpaceae). On river

banks and alluvial plains in south-western Tasmania, *N. cunninghamii* occurs with *Lagarostrobos franklinii* (Podocarpaceae), and occurs with *Athrotaxis selaginoides* (Taxodiaceae) in western Tasmania, usually above 600 m elevation (described by Kirkpatrick 1977; Jarman *et al.* 1984).

Although there has been extensive eco-physiological study of *N. cunninghamii*, primarily in Victoria (Howard 1970, 1973), little is known of the other rainforest tree species or of the rainforest dynamics. *A. moschatum* has been described as more shade-tolerant than *N. cunninghamii* on the basis of its ability to regenerate under the forest canopy (Gilbert 1959). The dominance of climax forest on equable sites by the supposedly shade-intolerant *N. cunninghamii* represents an anomaly in terms of forest dynamics, both with respect to the closely related forests of Chile and New Zealand, and to the temperate mesic forests of the northern hemisphere. The present study examines the comparative light requirements of the major canopy species occurring with *N. cunninghamii* in Tasmanian rainforest and discusses the results in terms of forest dynamics.

Methods

Instantaneous photosynthetic responses to light

Two-year-old seedlings of each species were grown in contrasting light regimes in a glasshouse from August to December 1984. Eight plants of each species were grown in (a) full sunlight and (b) shaded from direct sunlight so that maximum unfiltered diffuse light was 30–80 $\mu\text{mol quanta m}^{-2} \text{s}^{-1}$. The daily maximum temperatures in the glasshouse for the 7 weeks prior to and during measurement ranged from 23°C to 31°C with a mean maximum daily temperature of 26°C. Leaf temperature varied by less than 1°C between treatments. Another group of plants was grown for 12 months outdoors in a bush-house under a eucalypt woodland canopy of approximately 60% canopy cover (measured by hemispherical photograph). The daily maximum temperature for the 7 weeks prior to and during measurement ranged from 11°C to 24°C with a mean maximum daily temperature of 16°C.

An open gas exchange system incorporating Series 225 infra-red gas analyser (Analytical Development Co. Ltd) was used to measure the net rate of CO_2 assimilation at different photon flux densities (PFD). The light source was a Sylvania 400 W metalarc lamp. Photosynthetically active radiation was measured with a LiCor LI-185 meter with quantum sensor. Light intensity reaching the leaf was varied using a combination of distance from the light source and shade cloth screens. Humidified air (317–340 parts 10^{-6}CO_2) was passed through a leaf chamber of cross-sectional area 10.0 cm^2 at a flow rate of 0.4 l min^{-1} . Moisture was condensed from the gas lines prior to analysis. CO_2 differentials were recorded on a chart. Responses were first measured at 750 $\mu\text{mol quanta m}^{-2} \text{s}^{-1}$ and subsequently at decreasing light intensities. Responses at PFD greater than 750 $\mu\text{mol quanta m}^{-2} \text{s}^{-1}$ were measured last because of potential photoinhibition at high PFD. Leaf temperature was monitored and maintained at $21 \pm 1^\circ\text{C}$. Temperatures of 18–23°C were optimal for all species. Leaf photosynthesis was measured only on young fully expanded leaves of *N. cunninghamii*, *A. moschatum*, *E. lucida* and *P. aspleniifolius* (phylloclades). *A. selaginoides* and *L. franklinii* have small leaves, the latter closely appressed to the stem. Photosynthesis was measured in shoot sections rather than individual leaves of these plants. Total replicates were 5–8, with one leaf (or phylloclade or shoot) per plant. Foliage area was measured photometrically and varied in the range 3–10 cm^2 and leaf (or phylloclade) dry weight measured. The leaf area value of *A. selaginoides* used in the calculation of photosynthetic rate was taken from the projected area of the whole shoot in the leaf chamber. This was considered to be more accurate than total leaf area due to self-shading and leaf curvature. The leaf area and dry weight values used in the calculation of the specific leaf area for *A. selaginoides* involved measurement of the total area and dry weight of individual leaves. A similar inaccuracy occurs for *L. franklinii* though to a lesser extent. The projected area of the shoot was used for this species, and the dry weight of the entire shoot was used in calculation of the specific leaf area. Leaf chlorophylls of *N. cunninghamii*, *A. moschatum*, *E. lucida* and *P. aspleniifolius* were extracted in 80% (v/v) acetone and estimated spectrophotometrically.

Growth responses to light

Total plant response to light was examined by growing seedlings of the same species in four light intensities in a controlled environment growth cabinet. The light source consisted of a 400 W metalarc lamp, eight 40 W fluorescent lamps and four 60 W incandescent lamps for a 12 h photoperiod. Day temperature was 23°C and night temperature was 20°C. PFD was measured at the highest leaf of each seedling. The mean photon flux density of the treatments were 6, 48, 102 and 180 $\mu\text{mol quanta m}^{-2} \text{s}^{-1}$ giving daily photon fluxes of 0.26, 2.07, 4.40 and 7.78 $\text{mol quanta m}^{-2} \text{day}^{-1}$ respectively, with no significant difference between species within treatments. Leaf temperature between treatments varied by less than 1°C. Ten seedlings of each species were transferred from the glasshouse to each treatment and harvested after 35 days. Some deaths occurred and final replicates were 5–10 seedlings. Ten seedlings were harvested at the beginning of the experiment. Dry weight of each seedling was measured at each harvest. Mean relative growth rate was calculated as: $(\ln W_2 - \ln W_1) / (t_2 - t_1)$, where W_1 and W_2 are initial and final dry weights respectively, and $t_2 - t_1$ is the period of growth.

Results

Some details of the light-dependence curves (Figs. 1, 2) and leaf characteristics are summarized in Table 1. All species acclimated to shade conditions during growth by decreasing the dark respiration rate (DRR), instantaneous light compensation point (LCP) (except *P. aspleniifolius*) and light saturation point (LSP), and increasing the specific leaf area (SLA). Apparent quantum yield did not differ significantly between light regimes or species [$0.07 \pm 0.001 \text{ mol CO}_2. (\text{mol quanta})^{-1}$]. There were no indications of photoinhibition at incident PFD of up to 2500 $\mu\text{mol quanta m}^{-2} \text{s}^{-1}$.

The maximum rate of photosynthesis (P_{max}) of *A. moschatum* was unchanged by the light conditions during growth whereas *N. cunninghamii*, *E. lucida*, *P. aspleniifolius* and *L. franklinii* had lower P_{max} values in shade-raised seedlings ($P < 0.05$). *A. moschatum* had a low LCP and DRR in shade-grown plants and a low P_{max} in sun-raised plants compared with other species. Sun-raised plants were bleached, and had low chlorophyll values (Table 1). The SLA increased in shade conditions, with shade-raised *A. moschatum* having the highest SLA, and subsequently the highest chlorophyll/leaf

TABLE 1. Leaf characteristics and details from the light-dependence curves

Species	Growth light regime	SLA ($\text{m}^2 \text{g}^{-1} \times 10^{-3}$)		Chlorophyll (mg m^{-2})		(mg g^{-1} leaf dry weight)*	DRR ($\mu\text{mol m}^{-2} \text{s}^{-1}$)		LCP ($\mu\text{mol quanta m}^{-2} \text{s}^{-1}$)		LSP ($\mu\text{mol g}^{-1}$ leaf dry weight $\text{s}^{-1} \times 10^{-4}$)	
		mean	s.e.m.	mean	s.e.m.		mean	s.e.m.	mean	s.e.m.	mean	s.e.m.
<i>N. cunninghamii</i>	sun	131	7	383	24	50	1.73	0.12	27	3	8.10	0.61
	shade	149	8	406	15	60	1.05	0.12	17	2	4.96	0.31
	bush-house	139	8	540	11	75	0.36	0.03	6	1	6.10	0.51
<i>E. lucida</i>	sun	104	5	429	23	45	2.03	0.20	30	1	7.09	1.02
	shade	185	9	381	22	70	0.83	0.20	15	3	4.92	0.40
	bush-house	144	12	453	11	65	0.45	0.06	8	1	3.70	0.38
<i>A. moschatum</i>	sun	132	5	298	29	39	1.60	0.16	21	1	5.17	0.63
	shade	229	7	537	31	123	0.41	0.16	5	2	9.63	0.70
	bush-house	172	10	380	20	65	0.42	0.06	7	1	4.91	0.40
<i>P. aspleniifolius</i>	sun	41	4	484	47	20	0.87	0.15	11	2	2.01	0.17
	shade	115	6	618	54	71	0.92	0.16	13	2	4.24	0.32
	bush-house	106	2	555	40	58	0.41	0.05	7	1	3.75	0.35
<i>A. selaginoides</i>	sun†	24					1.61		20		1.07	
	shade	52	3				0.48	0.06	6	1	2.75	0.45
	bush-house	38	1				0.80	0.15	9	1	2.30	0.09
<i>L. franklinii</i>	sun	42	1				1.00	0.01	17	1	2.40	0.27
	shade	60	3				0.45	0.11	8	2	1.84	0.31
	bush-house	64	4				0.83	0.05	13	1	2.22	0.04

Each value is the mean of 5–8 replicates, with standard errors where appropriate.

*Calculated from the SLA; †Measurements are from only one plant.

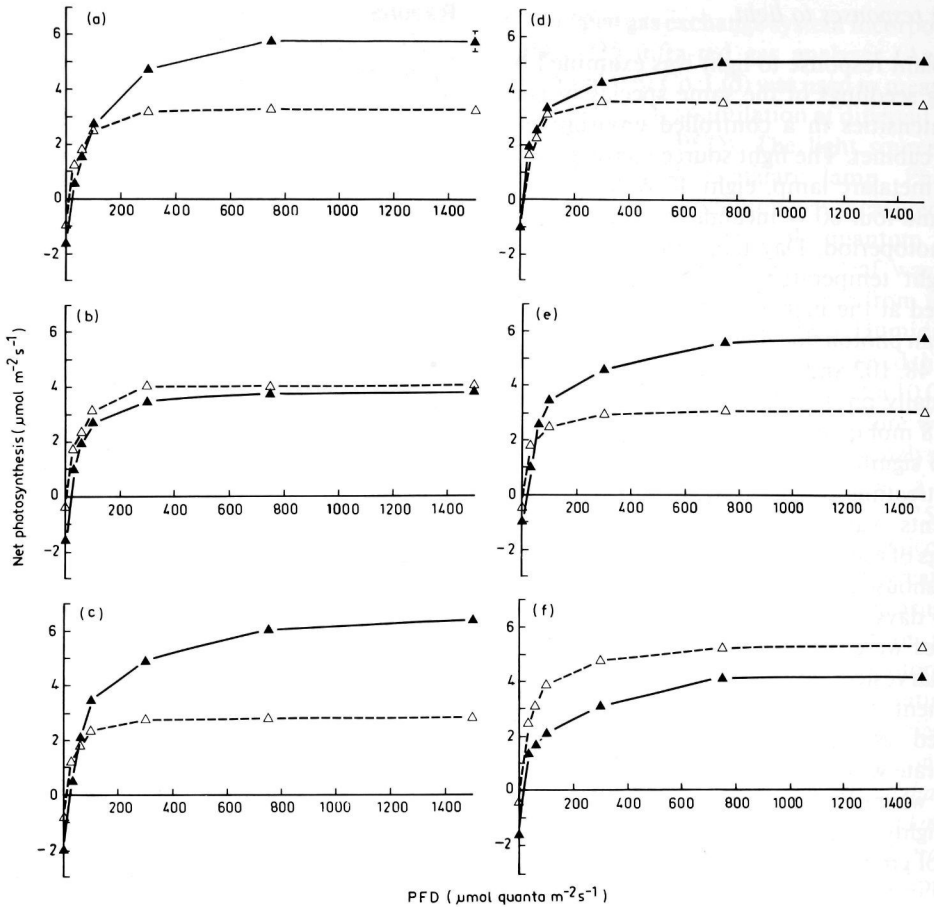


FIG. 1. Light-dependence curves of photosynthesis of seedlings grown in full sunlight (▲) and shade (△) in a glasshouse: (a) *N. cunninghamii*, (b) *A. moschatum*, (c) *E. lucida*, (d) *P. asplenifolius*, (e) *L. franklinii*, (f) *A. selaginoides*. Each point is the mean of 5–8 replicates. The largest standard error is shown on each curve, where it is greater than 0.25.

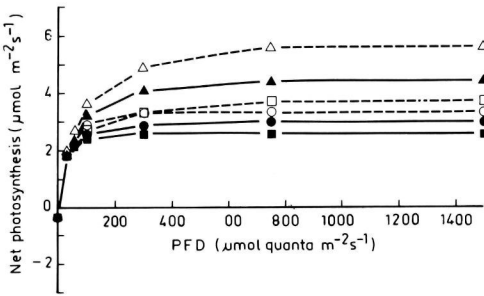


FIG. 2. Light-dependence curves of photosynthesis of seedlings grown outdoors under a woodland canopy: *N. cunninghamii* (▲); *A. moschatum* (●); *E. lucida* (■); *P. asplenifolius* (○); *L. franklinii* (□); *A. selaginoides* (△). Each point is the mean of five replicates. Standard errors are less than 0.25.

dry weight ratio and $P_{\max}$ on a leaf dry weight basis (Table 1). *A. moschatum* showed less acclimation when raised in a bush-house. These plants had a lower SLA, chlorophyll

content and $P_{\max}$ ($P < 0.05$) than shade-raised plants (Table 1).

N. cunninghamii, *E. lucida*, *P. asplenifolius* and *L. franklinii* had a lower $P_{\max}$ in shade-raised plants than sun-raised plants (Fig. 1; *N. cunninghamii* 57%, *E. lucida* 43%, *P. asplenifolius* 69% and *L. franklinii* 53%). Shade-raised *N. cunninghamii*, *E. lucida* and *L. franklinii* plants had a lower LCP and DRR than sun-raised plants (Table 1; $P < 0.05$). In *P. asplenifolius* these values were the same for sun and shade-raised plants (Table 1). The LCP and DRR of *N. cunninghamii*, *E. lucida* and *P. asplenifolius* were not as low as those of *A. moschatum* and they showed relatively less change in chlorophyll per unit leaf area in different light regimes (Table 1). *N. cunninghamii*, *E. lucida* and *L. franklinii* had their highest $P_{\max}$ on a dry weight basis in sun-raised plants (Table 1).

P. aspleniifolius had a higher $P_{\max}$, on a leaf area basis, in sun-raised plants (Fig. 1d), though a low SLA in these plants reduced the $P_{\max}$ on a leaf dry weight basis (Table 1). Seedlings of *P. aspleniifolius* and *A. selaginoides* grown in full light in the glasshouse showed evidence of stress, possibly due to high light/high temperature interactions. One of the *A. selaginoides* seedlings died and three seedlings failed to produce sufficient growth for use in this experiment and therefore only one sun-raised plant of *A. selaginoides* was suitable for use. Several *P. aspleniifolius* seedlings also failed to produce new growth and the phylloclades produced on the remaining plants were small and thickened. Consequently results from these species grown in full sunlight may not reflect field performance under cooler and more variable temperatures.

The growth rate study (Fig. 3) indicated that *A. moschatum* had a significantly higher relative growth rate (RGR) at the low daily photon fluxes (DPF) than the other species, but at $7.78 \text{ mol quanta m}^{-2} \text{ day}^{-1}$ did not differ significantly from the RGR of *N. cunninghamii* at the 95% level. There was no significant difference between the RGR of species other than *A. moschatum* at the low DPF (0.26 and $2.07 \text{ mol quanta m}^{-2} \text{ day}^{-1}$). At higher DPF

levels *N. cunninghamii* had significantly higher RGR ($P < 0.05$) than did *A. selaginoides*, *L. franklinii* (4.40 and $7.78 \text{ mol quanta m}^{-2} \text{ day}^{-1}$), or *P. aspleniifolius* ($7.78 \text{ mol quanta m}^{-2} \text{ day}^{-1}$). The RGR of *N. cunninghamii* did not differ significantly from that of *E. lucida* at the 95% level at any of the light levels.

Discussion

A. moschatum had a low LCP in shade-raised plants and had the limited capacity to acclimate photosynthetically to high quantum flux densities that are typical of shade-plants as described by Boardman (1977) and Björkman (1981). *A. moschatum* is only clearly superior to the other species when grown in heavy shade (lower DRR, LCP, higher $P_{\max}$ and higher RGR). In plants raised in a bush-house, the $P_{\max}$ of *A. moschatum* did not differ at the 95% level from that of *E. lucida* or *P. aspleniifolius* on either a leaf area or dry weight basis. This suggests that the light environment of the bush-house was higher than the light saturation point of *A. moschatum*. In contrast, the responses of *E. lucida*, *L. franklinii* and *P. aspleniifolius* when raised in a bush-house were most similar to results from shade-raised

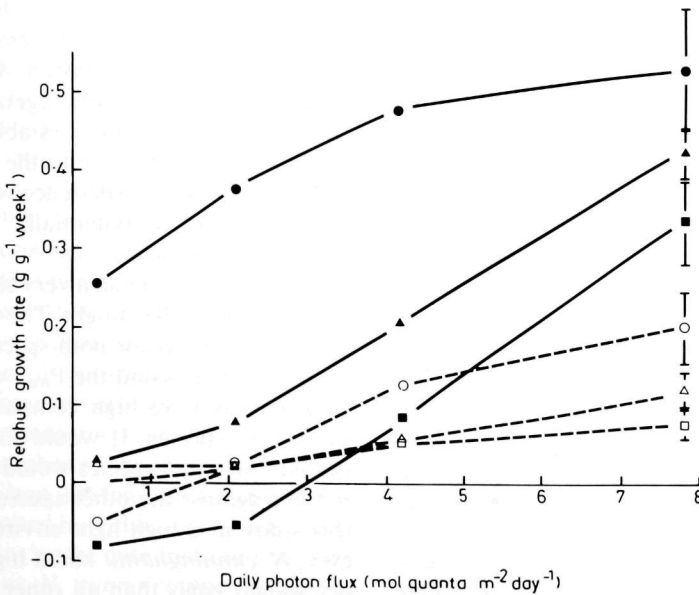


FIG. 3. Relative growth rates of seedlings grown under four light intensities in a controlled environment growth cabinet. *N. cunninghamii* (▲); *A. moschatum* (●); *E. lucida* (■); *P. aspleniifolius* (○); *L. franklinii* (□); *A. selaginoides* (△). Each point is the mean of 5–10 replicates, with the largest standard error shown for each species.

plants, and the response of *N. cunninghamii* plants raised in a bush-house was intermediate between that of sun and shade-raised plants. In *N. cunninghamii*, *E. lucida* and *P. aspleniifolius* the DRR and LCP was lower in plants raised in a bush-house than in sun-raised plants. It is not certain whether this was due entirely to characteristics of the light environment (variability and intensity of light) or can be related to different temperature and humidity conditions in the bush-house during growth.

N. cunninghamii, *E. lucida* and *P. aspleniifolius* do not separate clearly from one another on most of the major features of the sun and shade-raised plants. However, *N. cunninghamii* plants had a higher $P_{\max}$ on both a leaf area and dry weight basis when raised in the bush-house than did *E. lucida* and *P. aspleniifolius*. In addition, while the RGR of *N. cunninghamii* does not separate from that of any species in the lowest light conditions (except *A. moschatum*), at more moderate levels of shade ($7.78 \text{ mol quanta m}^{-2} \text{ day}^{-1}$) its RGR exceeds that of *L. franklinii*, *P. aspleniifolius* and *A. selaginoides*.

Although *P. aspleniifolius* had its highest $P_{\max}$ on a leaf area basis in the glasshouse, the low SLA in the high light/high temperature conditions experienced there reduced the $P_{\max}$ on a dry weight basis. However, these consistently high temperatures are not a feature of natural conditions in Tasmanian rainforest habitats. Therefore the optimum light environment for *P. aspleniifolius* is probably full sunlight.

The light-dependence curves for *A. selaginoides* suggest that seedlings raised in shade and a bush-house have a superior response to that of sun-raised seedlings. However, the growth-rate study indicates that *A. selaginoides* has a response typical of a shade-intolerant species, particularly in its response to the lowest light levels. This suggests that the light-dependence curve for the sun-raised seedling may reflect a high light/high temperature interaction which would not be experienced in natural conditions in Tasmania. This view is supported by observations in the field which indicate that *A. selaginoides* regenerates successfully only in relatively high light environments (Read & Hill, unpublished data).

All three gymnosperms had a low $P_{\max}$ on a dry weight basis due to low SLA values relative to the other species. These species also had a low leaf area ratio (unpublished data). These features would be expected to have an impact on the dynamics of *P. aspleniifolius* and *L. franklinii* (and probably *A. selaginoides*) since they acclimate to high photon flux densities and therefore are likely to reach peak photosynthesis in canopy gaps where they are competing with faster growing species. Although *L. franklinii* (and *A. selaginoides*) differed from the other species with high light saturation points (*N. cunninghamii*, *E. lucida* and *P. aspleniifolius*) in having a low DRR and LCP in shade-raised plants, the growth rate experiment suggests that the low leaf area ratio and SLA produce a growth LCP that does not differ from that of the other species with high light saturation points. The dynamics of forests containing these species are currently being studied.

A. moschatum was the most shade-tolerant of the species studied and therefore would be expected to regenerate under denser canopies than would the other species. However, predictions and explanations of natural regeneration patterns based on these results are limited by the effect of other environmental factors on establishment and growth. This is evident in the case of *A. moschatum* which regenerates in the rainforest primarily by basal sprouts from established stems (Johnston & Lacey 1983; Read & Hill 1985). While vegetative reproduction may further facilitate establishment under low light conditions due to the absence of the stresses commonly experienced during seedling establishment, it potentially limits spread through the rainforest.

N. cunninghamii had a very similar response to *E. lucida* in full sunlight. The maximum rate of photosynthesis for both species occurred in sun-raised plants and the $P_{\max}$ values on a dry weight basis were high compared with other sun-grown species. It would therefore be expected that these species would be most competitive against the other species examined in this study in a high light environment. However, *N. cunninghamii* had a higher $P_{\max}$ on a dry weight basis than all other species in the bush-house, suggesting that it is also very competitive in this intermediate light environment.

These features are in agreement with analysis of the spatial distribution of seedlings of these species in lowland rainforests, which indicates that regeneration is predominantly clumped in canopy gaps (Read & Hill 1985, unpublished data).

Explanations and predictions of successional trends in mesic forest on equable sites are usually correlated with the comparative shade-tolerance of the canopy species, with the most shade-tolerant canopy species dominating old, undisturbed forests (Jones 1945; Forcier 1975; Williamson 1975; Noble & Slatyer 1978). This contrasts with the situation in Tasmania where the cool temperate rainforest is commonly dominated by *N. cunninghamii*, which is regenerating continuously, even when it is co-occurring with the more shade-tolerant *A. moschatum* (Howard 1981; Read & Hill 1985). The Tasmanian rainforest also contrasts with the *Nothofagus*-dominated rainforests of Chile and New Zealand. Studies in these areas suggest that when *Nothofagus* occurs with shade-tolerant rainforest trees on equable sites it fails to regenerate sufficiently to maintain its dominance unless there is exogenous disturbance (Veblen & Ashton 1978; Veblen *et al.* 1980, 1981; Wardle 1983, 1984). Clearly factors other than shade-tolerance are important in determining the dominant status of *N. cunninghamii* in Tasmanian rainforest, at least with respect to *A. moschatum*. Shade-intolerant trees commonly have an advantage in reproductive strategies (Forcier 1975; Williamson 1975), microsite preferences (Billings 1938; Bormann 1953; Veblen *et al.* 1981) and relative growth rate (Bray 1956; Williamson 1975; Veblen *et al.* 1981) which facilitate early dominance on a disturbed site. If these advantages operate for *N. cunninghamii* at a small scale of disturbance such as canopy gaps created by the death of an old tree, then its dominance over *A. moschatum* is explicable. One clear difference between the Tasmanian cool temperate rainforest and that of Chile lies in the larger number of shade-tolerant canopy species in Chilean lowland and mid-altitude rainforest (six cited by Veblen & Ashton 1978). It may therefore be the case that the relatively shade-intolerant *N. cunninghamii* is able to dominate mature forest in Tasmania because of the lack of shade-tolerant species with effective establishment strategies.

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